



IILM UNIVERSITY

SUMMER INTERNSHIP PRESENTATION

Crop and Weed Detection

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Program & Semester: B.Tech CSE | 3rd Semester

Section: 2CSE33 | Batch: 2026–27

Organization Name: Uniconverge Technologies (P) Ltd. – upSkill Campus

Internship Overview

Organization: Uniconverge Technologies (P) Ltd. – upSkill Campus

Internship domain: Machine Learning / Python

Project: Crop and Weed Detection

Focus on computer vision and object detection for identifying crops and weeds.

The work covered the workflow from dataset preparation to model training and prediction.

Problem Statement

Manual crop and weed identification can be time-consuming and inconsistent.

An automated image-based approach can assist in locating and classifying crop and weed objects.

The project develops a learning-oriented object-detection workflow for agricultural images.

The implementation demonstrates practical application of Python, computer vision and machine learning.

Objectives

Dataset Preparation

Organize images, annotations and preprocessing steps required for object detection.

Model Development

Use a YOLO-based object-detection workflow for crop and weed detection.

Prediction

Run the trained model on images and interpret bounding-box detections.

Practical Learning

Connect Python, computer vision, data preparation and machine-learning concepts.

Documentation

Record implementation, challenges, outcomes and project learnings.

Technologies & Tools

Python — primary programming language

YOLO-based object detection workflow

Computer vision and image preprocessing

Dataset annotation and augmentation

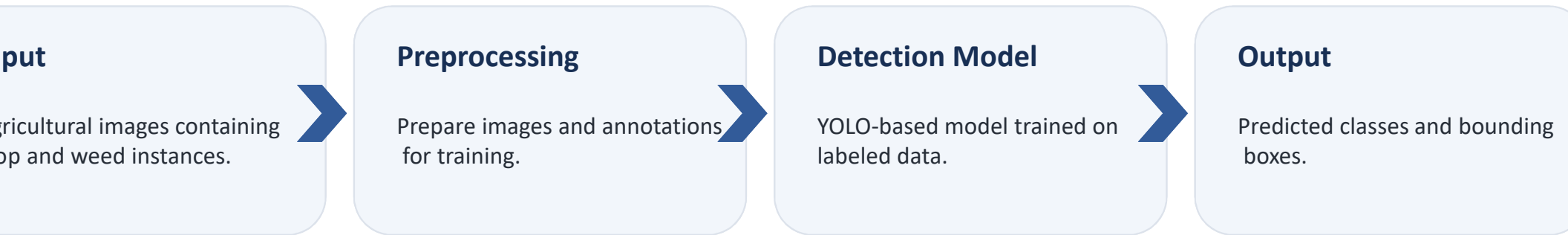
Bounding-box based detection

Model training and image prediction

Implementation Workflow



Project Architecture



Prepared images → trained detection model → visual predictions

Challenges & Solutions

Data Quality

Validate images and annotations so training data is consistent.

Model Understanding

Break training into stages and inspect outputs after each stage.

Generalization

Use suitable preprocessing and augmentation to increase data variety.

Debugging

Check file paths, labels and configuration systematically.

Learning Outcomes

- Improved understanding of Python-based machine-learning workflows.
- Learned practical stages of computer-vision object detection.
- Gained experience with dataset preparation, annotation and augmentation.
- Improved debugging by validating each stage of an ML pipeline.
- Learned to interpret predictions using bounding boxes and class labels.
- Improved technical documentation and presentation skills.

Conclusion & Future Scope

The project demonstrates an end-to-end crop and weed detection workflow using Python and object-detection concepts.

The internship provided practical exposure to machine learning, computer vision and structured project development.

Future work can use larger and more diverse agricultural datasets and evaluate model performance more extensively.

The workflow can be extended toward real-time or field-deployment scenarios after appropriate validation.

Internship Deliverables

Project Code

Crop and Weed Detection implementation maintained in GitHub.

Internship Report

Structured report documenting the project and learning outcomes.

Presentation

Professional summary for academic evaluation.

GitHub

Public repository prepared for submission.

Certificate

Internship completion certificate to be added when issued.

THANK YOU